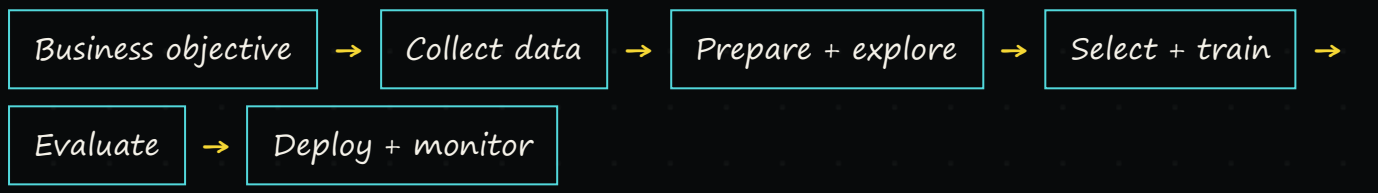
How Data Drives Decision Making in Machine Learning

Frame the business decision before gathering data. The ML workflow connects a measurable objective to collected, prepared, understood data, a selected model, evaluation, deployment, and continuing maintenance.

Original study notes based on Gwendolyn Stripling and Michael Abel, First edition, 2023. Book pp. 1–16. Source APIs, services, experiments, and model comparisons reflect the book's publication period. These are explanatory summaries, not verbatim reproductions or a claim that every source detail is retained.



Detailed Notes

Start with a decision, not an algorithm

Source: book pp. 1–5 / PDF p. 17

Identify who makes the decision, what action changes, and what measurable outcome matters. Translate a broad request ("use AI") into a problem statement: prediction target, unit of analysis, decision timing, users, and constraints.

Clarify whether the task is regression, classification, ranking, or another objective. Determine what happens if the model is wrong, how a human will use the prediction, and what a non-ML baseline already achieves. Data collection should follow the use case because irrelevant data cannot rescue an unclear question.

Enterprise ML workflow: decisions and data

Source: book pp. 1–10 / PDF p. 17

The workflow is iterative, not a one-way checklist:

1. Define business objective and success metric.
2. Collect appropriate, permissioned data.
3. Preprocess types, missing values, errors, and duplicates.
4. Explore distributions, relationships, and leakage.
5. Transform/select features using training data.
6. Choose an AutoML, SQL, low-code, or custom approach.
7. Train, evaluate, tune, and compare with a baseline.
8. Test on held-out data and operational conditions.
9. Deploy predictions through a usable service/workflow.
10. Monitor, maintain, and retrain when justified.

Data processing and problem definition may loop as discoveries change the question.

Model selection, tuning, and testing

Source: book pp. 11–14 / PDF p. 27

AutoML can automate candidate algorithms, feature processing, search, training, and evaluation, but the practitioner still sets the objective, data split, constraints, and acceptance bar. Testing means evaluating data and behavior not used to fit or tune the model.

Check that an offline metric relates to the business decision. Test integration, latency, throughput, permissions, and behavior on realistic new data. Save preprocessing and model together so serving uses the same transformations.

Deployment and maintenance

Source: book pp. 14–16 / PDF p. 30

Serving can be online (respond to a request) or batch (predict a collection on a schedule). Account for model availability, cost, data access, monitoring, and rollback.

Input distributions and relationships can change. Monitor data quality and outcome metrics; retrain or revise only when evidence warrants it. Version datasets, transformations, and models to diagnose a change. A model that once performed well is not permanently validated.

Revision Check

1. What decision will the prediction change?
2. What is the simplest baseline?
3. Which data might leak the answer?
4. How will you know the deployed model still helps?

The business question determines the data, target, metrics, and deployment. Start there, then choose the lightest workflow that can meet the measured need.

Source links refer to the supplied PDF. Every contents-page topic is mapped in coverage.md. Consult the book for complete implementations, figures, exercises, citations, and experiment evidence.